

Introduction

AI and ML at Work

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**Autonomous Cars and
Navigation**

AI and ML at Work



Autonomous Cars and Navigation



"Alexa", "Siri", "Cortana" etc.

AI and ML at Work

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Creativity: Generated Images

AI and ML at Work



Creativity: Generated Images



**Playing Games better than
Human**

Modern AI: End 2 End Driving

Modern AI: End 2 End Driving



Modern AI: End 2 End Driving



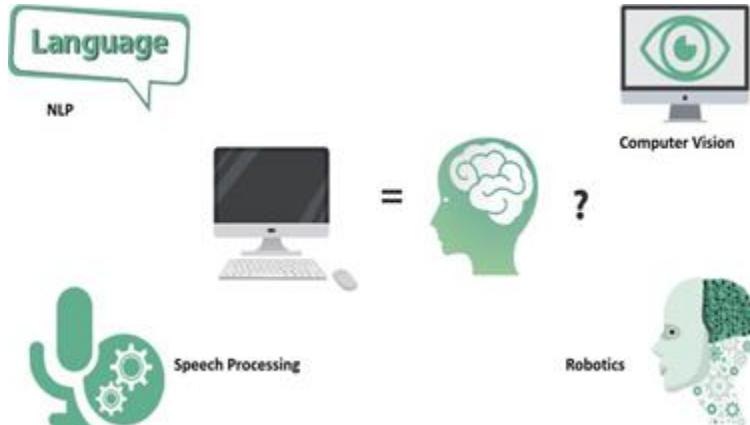
What is Modern AI and ML



Modern “AI”



Modern “AI”



A Simplified View

Demystifying ML

A Simple Question

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- ▶ How do we design a computational procedure?

A Simplified View of ML



Eg:
int function(intx[]) {
...
}

A Simple Question (cont.)

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- ▶ 0.99, 3.02, 9.00, 18.98, 33.01, ... What next?

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Consider a series of 2D
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▶ $(1,3), (2,6), (3,9),$
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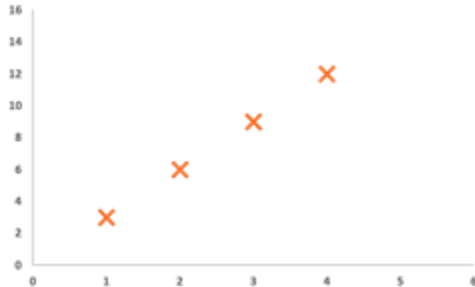
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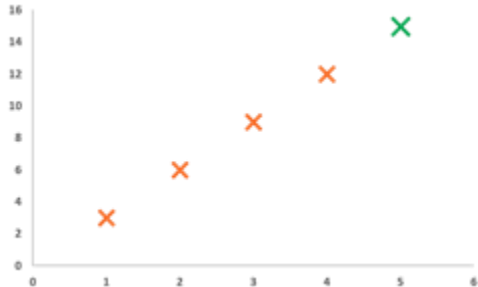
▶ (1,3), (2,6), (3,9), (4,12),

....

▶ What is the next point?

▶ (x, 3x) or

▶ Function: $y=f(x)=3x$



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- ▶ When numbers are “uncertain”
 - Noise in measurements
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- ▶ When numbers are not just “simple numbers”
 - 2D points, 3D points
 - 100 Dimensional points

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► When the function is complex or function nature is unknown

- Simple linear functions are easy to guess.
 - Eg. $f(x) = \omega_1 x + \omega_2$
- Finding “best” parameters/coefficients can be hard.
 - What is the best “ ω ” that suits the data?

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- ▶ Both mine and your solutions can be right?

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 - Does a patient with 7.7 have anemia?
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 - Why 10? Why not 12?
 - Multiple solutions. Both works well now. Future?

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 - The Data
 - Data + interpretations $(x, y) = (\text{sample}, \text{label})$
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▶ Who gives functional form

- Most problems need complex functions
(Note that simple “Linear” solutions are also good in many cases.)

Closer Look..

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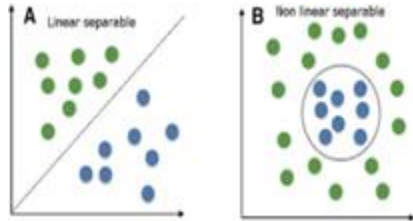
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- ▶ How to find the “optimal” parameters?
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 - Training
 - Computing
- ▶ How do we expect that it will work well in the future?

“Classification”: A Popular Problem

- ▶ **Example:**
 - Given medical records, predict presence of Malaria
- ▶ **Data:** A set of samples $\{ X \}$ labeled by experts.
- ▶ **Performance:** Predict accurately on unseen data
- ▶ $\{0,1\}$ classification
 - “Yes” or “No”
 - Yes if $f(x) > 0$
- ▶ **Multiclass classification**
- ▶ **Many more variants**



Problem Space



**talent
sprint**
Part of Accenture

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HYDRABAD

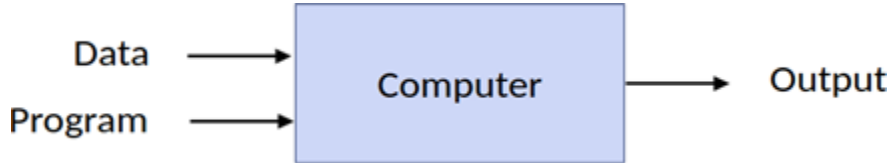
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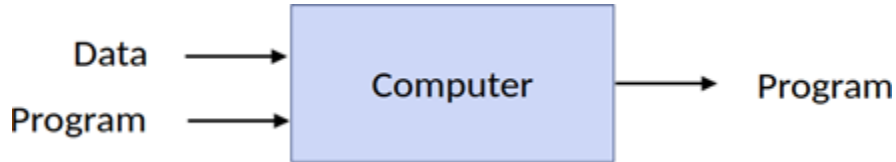
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- ▶ **End to End:** Can we learn y directly from I

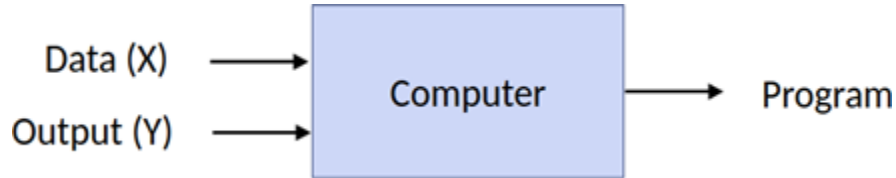
Traditional Programming



Machine Learning



Machine Learning



Thanks!!
Questions?